

Chapter 5

The Relational Data Model and Relational Database Constraints

1

Chapter Outline

- Relational Model Concepts
- Relational Model Constraints and Relational Database Schemas
- Update Operations and Dealing with Constraint Violations

2

Informal Definitions

- A **relational database** is a collection of relations.
- Each **relation** looks like a **table** of values.
 - Each column corresponds to an **attribute**.
 - A relation typically contains a set of rows, and each row contains a value for each attribute.
 - In the formal relational model, the rows in relations are called **tuples**.

3

A Relational Database Example

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

4

Relation Schema

- A **relation schema** R , denoted by $R(A_1, A_2, \dots, A_n)$, is made up of a relation name R and a list of attributes $A_1, A_2, \dots, A_n$.
- Example:
 $CUSTOMER(Cust_id, Cust_name, Address, Phone_no)$
 - $CUSTOMER$ is the relation name.
 - Attributes are: $Cust_id, Cust_name, Address, Phone_no$
- Each attribute has a **domain** which can be defined as a data type, a format, or a set of possible (i.e., valid) values.
 - For example, the domain of $Cust_id$ could be a set of 6 digit numbers.
- The **degree** (or **arity**) of a relation is number of attributes in its schema.

5

Relation

- An **instance** (or **state**) of a relation schema is called a **relation** (or **relation state**, **relation instance**).
- If r is a relation of the relation schema $R(A_1, A_2, \dots, A_n)$, it is denoted by $r(R)$.
- $r(R)$ is a set of tuples:
 - $r(R) = \{t_1, t_2, \dots, t_m\}$ where each t_i is an **n-tuple**.
 - Each **n-tuple** t_i is an ordered list of n values:
 $t_i = \langle v_1, v_2, \dots, v_n \rangle$, where each v_j is an element of $\text{domain}(A_j)$ or is a NULL value.
 - Each tuple in the $CUSTOMER$ relation is a 4-tuple and consists of four values; for example, $\langle 632895, 'John Smith', '101 Main St. Atlanta, GA 30332', '404-894-2000' \rangle$

6

Example of a Relation

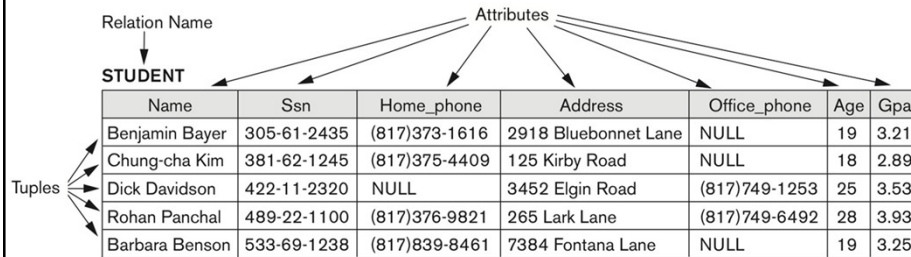


Figure 5.1 The attributes and tuples of a relation STUDENT.

7

Characteristics of Relations

- A relation is defined as a set of tuples, hence the order of tuples within a relation is not important.
- An n-tuple is an ordered list of n attribute values, hence the order of attribute values within a tuple is important.

8

Figure 5.1 The attributes and tuples of a relation STUDENT.

Relation Name		Attributes						
STUDENT								
		Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Tuples		Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
		Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
		Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
		Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
		Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Figure 5.2 The relation STUDENT from Figure 5.1 with a different order of tuples.

STUDENT						
Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21

9

Values in the Tuples

- A composite attribute is not allowed in a relation schema: we keep only the simple component attributes.
- Each tuple cannot have multiple values for any attribute. So, a multivalued attribute of an entity type must be represented by a separate relation schema (together with a key).
 - For example, DEPARTMENT_LOCATIONS(Dnumber, Location) in addition to DEPARTMENT(Dname, Dnumber, Mgr_ssn, Mgr_Start_Date)
- As a result, each tuple has a single atomic value for each attribute.
- A special **NULL** value is used in a tuple for an attribute if the attribute value is unknown or inapplicable.
 - For example, spouse name for a single employee, phone number of an employee not known at the time of inserting his/her tuple, etc.

10

Values in the Tuples (cont'd)

- Notation:
 - We refer to the **value of an attribute A_i in a tuple t** by:
 - $t[A_i]$ or $t.A_i$
 - This is the value v_i of attribute A_i in tuple $t = \langle v_1, v_2, \dots, v_n \rangle$.
 - Similarly, $t[A_u, A_v, A_w]$ refers to a **subtuple** of t containing only the values of attributes A_u, A_v, A_w in t .

11

Keys of a Relation Schema

- A **superkey** of R :
 - is a set of attributes SK of R with the following condition:
 - No two tuples in any relation $r(R)$ will have the same value for SK .
 - That is, for any two distinct tuples t_1 and t_2 in $r(R)$, $t_1[SK] \neq t_2[SK]$.
 - This condition must hold in any relation $r(R)$.
- A **key** of R :
 - A key is a "minimal" superkey.
 - That is, a superkey K is a key if removal of any attribute from K results in a set of attributes that is not a superkey (i.e., does not possess the **uniqueness property** of superkey).
 - **Thus, a key is a minimal set of attributes for which different tuples should have different combination of values.**
 - Any key is a superkey (but not vice versa).
 - Any set of attributes that includes a key is a superkey.
 - **A superset of a key is a superkey.**

12

Keys of a Relation Schema (cont'd)

- A relation schema may have more than one key. In this case, each of the keys is called a **candidate key**.
- Example: Consider the CAR relation schema:
CAR(License_number, engine_serial_number, Make, Model, Year)
 - **CAR** has two keys: License_number and Engine_serial_number

13

Keys of a Relation Schema (cont'd)

- If a relation has several **candidate keys**, one is chosen arbitrarily to be the **primary key**, and other candidate keys are designated as **unique keys** (aka **secondary keys**).
 - The primary key value is used to uniquely identify each tuple in a relation.
 - The **primary key attributes** (i.e., the attributes belong to the primary key) **are underlined**, but unique key attributes are not underlined.

CAR

<u>License_number</u>	<u>Engine_serial_number</u>	Make	Model	Year
Texas ABC-739	A69352	Ford	Mustang	02
Florida TVP-347	B43696	Oldsmobile	Cutlass	05
New York MPO-22	X83554	Oldsmobile	Delta	01
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	04
Texas RSK-629	U028365	Jaguar	XJS	04

Figure 5.4 The CAR relation, with two candidate keys: License_number and Engine_serial_number.

14

How to Specify the Primary and Unique Keys in SQL?

```
CREATE TABLE PROJECT
(Pnumber INT,
 Pname VARCHAR(15) NOT NULL,
 Plocation VARCHAR(15),
 Dnum INT NOT NULL DEFAULT 10,
PRIMARY KEY (Pnumber),
UNIQUE (Pname),
FOREIGN KEY (Dnum) REFERENCES
DEPARTMENT(Dnumber) ON DELETE SET
DEFAULT ON UPDATE CASCADE);
```

15

Relational Database Schema

- **A relational database schema is a set of relation schemas and a set of constraints.**
 - For example, a relational database schema can be specified as: $S = \{R_1, R_2, \dots, R_n\}$
 - S is the **relational database schema**.
 - $R_1, R_2, \dots, R_n$ are the individual **relation schemas** within S.
- **A relational database is an instance (or state) of a relational database schema:**
 - A set of relations, containing one instance (or state) for each relation schema.

16

COMPANY Relational Database Schema

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	-----	-------	---------	-----	--------	-----------	-----

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
-------	---------	---------	----------------

DEPT_LOCATIONS

Dnumber	Dlocation
---------	-----------

PROJECT

Pname	Pnumber	Plocation	Dnum
-------	---------	-----------	------

WORKS_ON

Essn	Pno	Hours
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DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
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Figure 5.5 Schema diagram for the COMPANY relational database schema.

1 /

A Relational Database: an Instance (or State) of the COMPANY Database Schema

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
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Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
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5	Bellaire
5	Sugarland
5	Houston

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123456789	2	7.5
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453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
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ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
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333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

18

Relational Integrity Constraints

- **Constraints are conditions** that must hold on all the relations.
- There are three main types of constraints in the relational model:
 - **Key constraint**
 - **Entity integrity constraint**
 - **Referential integrity constraint** (aka Foreign key constraint)
- Another implicit constraint is the **domain constraint**.
 - Every value in a tuple must be from the domain of its attribute (or it could be **NULL** if allowed for that attribute).

19

Key and Entity Integrity Constraints

- **Key Constraint:** Every candidate key value must be unique for each tuple in each relation of that relation schema.
 - In other words, different tuples in a relation should have different values for each candidate key.
 - NULL value may be allowed for a unique key (aka secondary) because NULL values are considered to be different from each other.
- **Entity Integrity Constraint :** The primary key of each relation schema R cannot have a NULL value in any tuple of $r(R)$.
 - This is because primary key values are used to identify individual tuples within the relation.
 - If the primary key is composed of multiple attributes, NULL is not allowed for any of these attributes.

20

Referential Integrity Constraint (aka Foreign Key Constraint)

- A Referential Integrity Constraint involves two relations: a **referencing relation** and a **referenced relation**.
- If the primary key of a relation R_2 is also included in a relation R_1 ,
 - R_1 is called a **referencing relation**,
 - R_2 is called a **referenced relation**, and
 - the primary key of R_2 that is contained in R_1 is called a **Foreign Key** of R_1 .
 - A foreign key and its corresponding primary key may have different (attribute) names.
- If the **primary key** of the referenced relation R_2 is denoted by PK , and the **foreign key** of the referencing relation R_1 is denoted by FK , then a tuple t_1 in R_1 is said to **reference** a tuple t_2 in R_2 if $t_1[FK] = t_2[PK]$.

21

How to Specify a Foreign Key in SQL?

```
CREATE TABLE PROJECT
(Pnumber INT,
 Pname VARCHAR(15) NOT NULL,
 Plocation VARCHAR(15),
 Dnum INT NOT NULL DEFAULT 10,
 PRIMARY KEY (Pnumber),
 UNIQUE (Pname),
 FOREIGN KEY (Dnum) REFERENCES
 DEPARTMENT(Dnumber) ON DELETE SET
 DEFAULT ON UPDATE CASCADE);
```

22

Referential Integrity Constraint (cont'd)

- Referential Integrity Constraint: The value of the foreign key *FK* of the referencing relation R_1 can be either:
 - An existing value of the primary key *PK* in the referenced relation R_2 , or
 - NULL (if allowed).
- A referential integrity constraint can be displayed in the relational database schema diagram as a directed arc from $R_1.FK$ to $R_2.PK$ (or R_2 if *FK* attribute names are the same as those of *PK*).

23

Referential Integrity Constraints for COMPANY Database

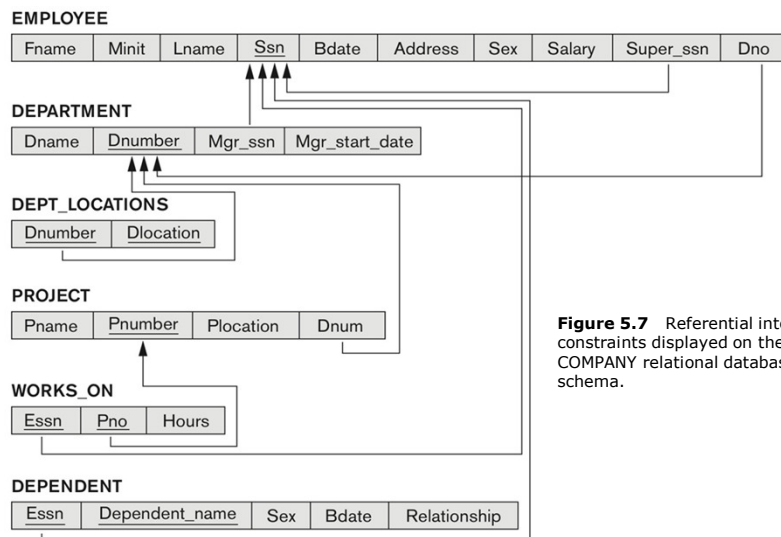


Figure 5.7 Referential integrity constraints displayed on the COMPANY relational database schema.

24

Other Types of Constraints

- **Semantic Integrity Constraints:**
 - based on application semantics and cannot be enforced by (the basic domain and integrity constraints of) the relational model.
 - Examples:
 - The maximum number of hours of each employee for all the projects he/she works on is 56 hours per week.
 - The salary of an employee cannot be higher than that of his/her supervisor.
- A **constraint specification language** may be used to express them.
- SQL-99 (aka SQL3) allows **assertions** and **triggers** to express them.

25

Update Operations on Relations

- **INSERT, DELETE, or UPDATE (or MODIFY) a tuple.**
- Integrity constraints should not be violated by the update operations. In case of integrity violation, several actions can be taken:
 1. Cancel the operation that causes the violation.
 2. Perform the operation, but inform the user of the violation.
 3. Trigger additional updates so the violation is corrected.
 4. Execute a user-defined function (or procedure).

26

Possible Violations for Each Update Operation

- **INSERT may violate any of the following constraints:**
 - Domain constraint:
 - if one of the attribute values provided for the new tuple is not in the specified attribute domain.
 - Key constraint:
 - if the key value of the new tuple already exists as the key value of another tuple in the relation.
 - Entity integrity constraint:
 - if the primary key value is NULL in the new tuple.
 - Referential integrity constraint:
 - if a foreign key value (which is not NULL) in the new tuple in the referencing relation does not exist as a primary key value in the referenced relation.

27

Possible Violations for Each Update Operation (cont'd)

- **DELETE may violate only the referential integrity constraint:**
 - If the primary key value of the tuple being deleted is referenced by other tuples in the database, it can be remedied by certain actions (see the chapters on SQL for more details):
 - RESTRICT option: reject the deletion.
 - SET NULL option: set the foreign key values of the referencing tuples to NULL.
 - SET DEFAULT option: set the foreign key values of the referencing tuples to a specified default value.
 - CASCADE option: delete the referencing tuples.
 - One of the above options must be specified for each foreign key during database design.

28

How to Specify a Foreign Key in SQL?

```
CREATE TABLE PROJECT
(Pnumber INT,
 Pname VARCHAR(15) NOT NULL,
 Plocation VARCHAR(15),
 Dnum INT NOT NULL DEFAULT 10,
 PRIMARY KEY (Pnumber),
 UNIQUE (Pname),
 FOREIGN KEY (Dnum) REFERENCES
 DEPARTMENT(Dnumber) ON DELETE SET
 DEFAULT ON UPDATE CASCADE);
```

29

Possible Violations for Each Update Operation (cont'd)

- **UPDATE** may violate the domain constraint or the **NOT NULL** constraint (if specified) on an attribute being modified.
- Any of the other constraints may also be violated, depending on the attribute being updated:
 - **Updating the primary key (PK) value:**
 - **RESTRICT** option: reject the update if a referencing tuple exists.
 - **SET NULL** option: set the foreign key values of the referencing tuples to NULL (if allowed).
 - **SET DEFAULT** option: set the foreign key values of the referencing tuples to a specified default value.
 - **CASCADE** option: propagate the new primary key value into the foreign key of the referencing tuples.
 - **Updating a foreign key (FK) value:**
 - May violate the referential integrity constraint.

30